

Doing Bayesian Data Analysis

Chapter 7

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Difficulty

- In the usual case of continuous parameters, the integral of the marginal likelihood can be impossible to solve analytically:

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)} = \frac{p(D|\theta)p(\theta)}{\int p(D|\theta)p(\theta)d\theta}$$

- MCMC approximations generate representative parameter-value combinations from the posterior distribution of complex models without computing the integral in Bayes' rule.

Bayesian and Computing Power

- This chapter introduces the methods we will use for producing accurate approximations to Bayesian posterior distributions for realistic applications.
- The class of methods is called Markov chain Monte Carlo (MCMC).
- It is MCMC algorithms and software, along with fast computer hardware, that allow us to do Bayesian data analysis for realistic applications that would have been effectively *impossible* 30 years ago.

← Which year did the author write this statement?

Curse of Dimensionality

- Grid approximation is one approach to addressing the situations when posterior distributions are not analytically solvable.
- If we set up a comb on each parameter that has 1,000 values, then the six-dimensional parameter space has $1,000^6 = 1,000,000,000,000,000,000$ combinations of parameter values.
- Too many for *any* computer to evaluate.

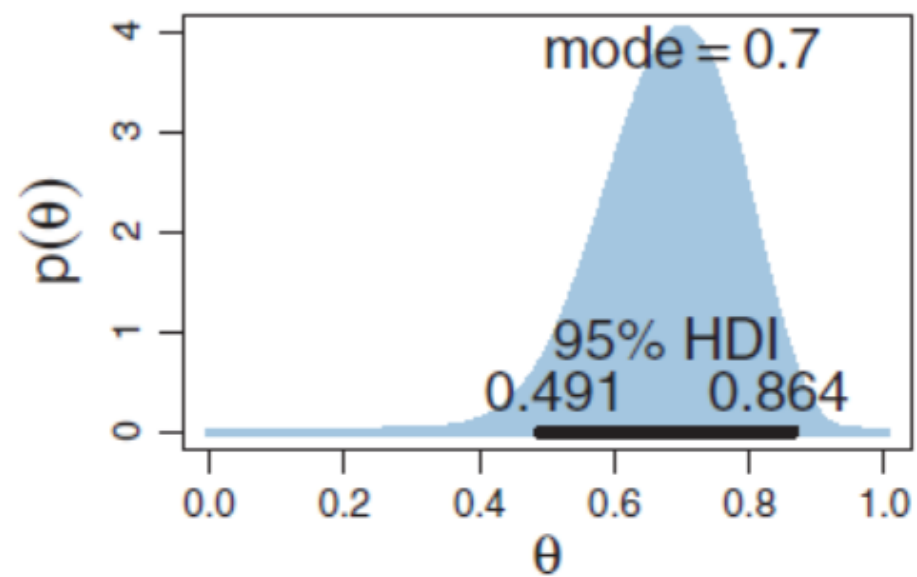
Sampling is Easier

- All the MCMC method really demands is the prior and likelihood can be easily computed up to a multiplicative constant.
- The method does not require evaluating the difficult integral in the denominator of Bayes' rule.
- The posterior is estimated by *randomly generating a lot of θ values* from it. Therefore, this approach is called a Monte Carlo (MC) method.

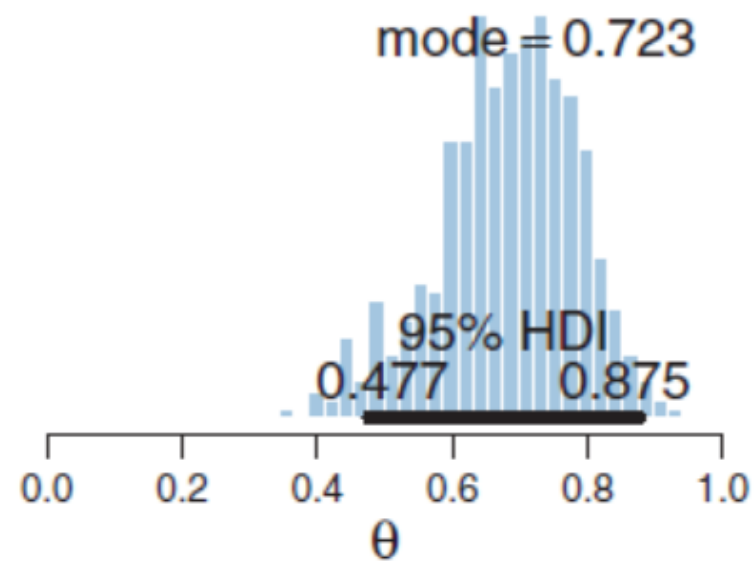
Sampling is Useful

- The concept of representing a distribution by a large representative sample is not new: Applied routinely in polls and surveys.
- By randomly sampling a subset of people from a population, we estimate the underlying tendencies in the entire population. The larger, the better.
- What is new is that the population from which we are sampling is a mathematically defined distribution.

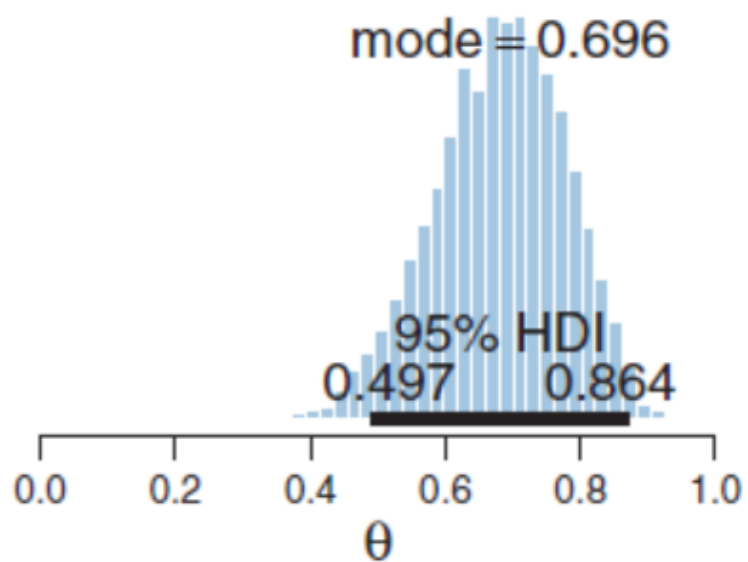
Exact Distribution



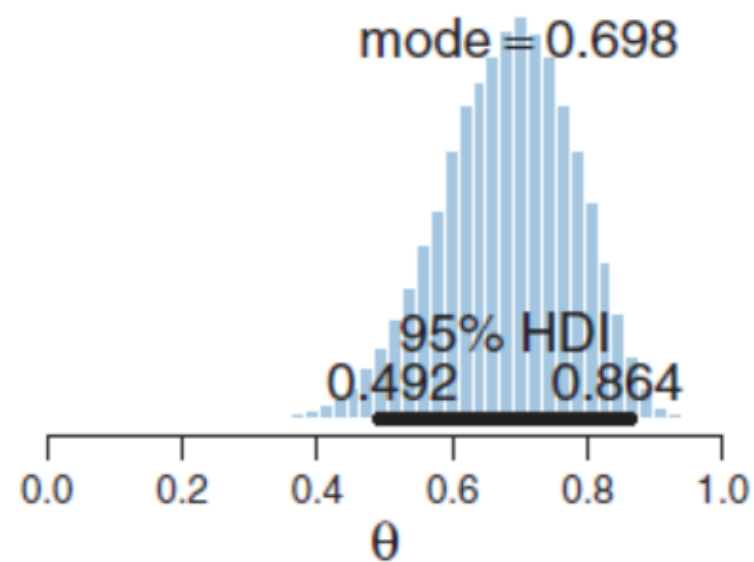
Sample N = 500



Sample N = 5000



Sample N = 50000



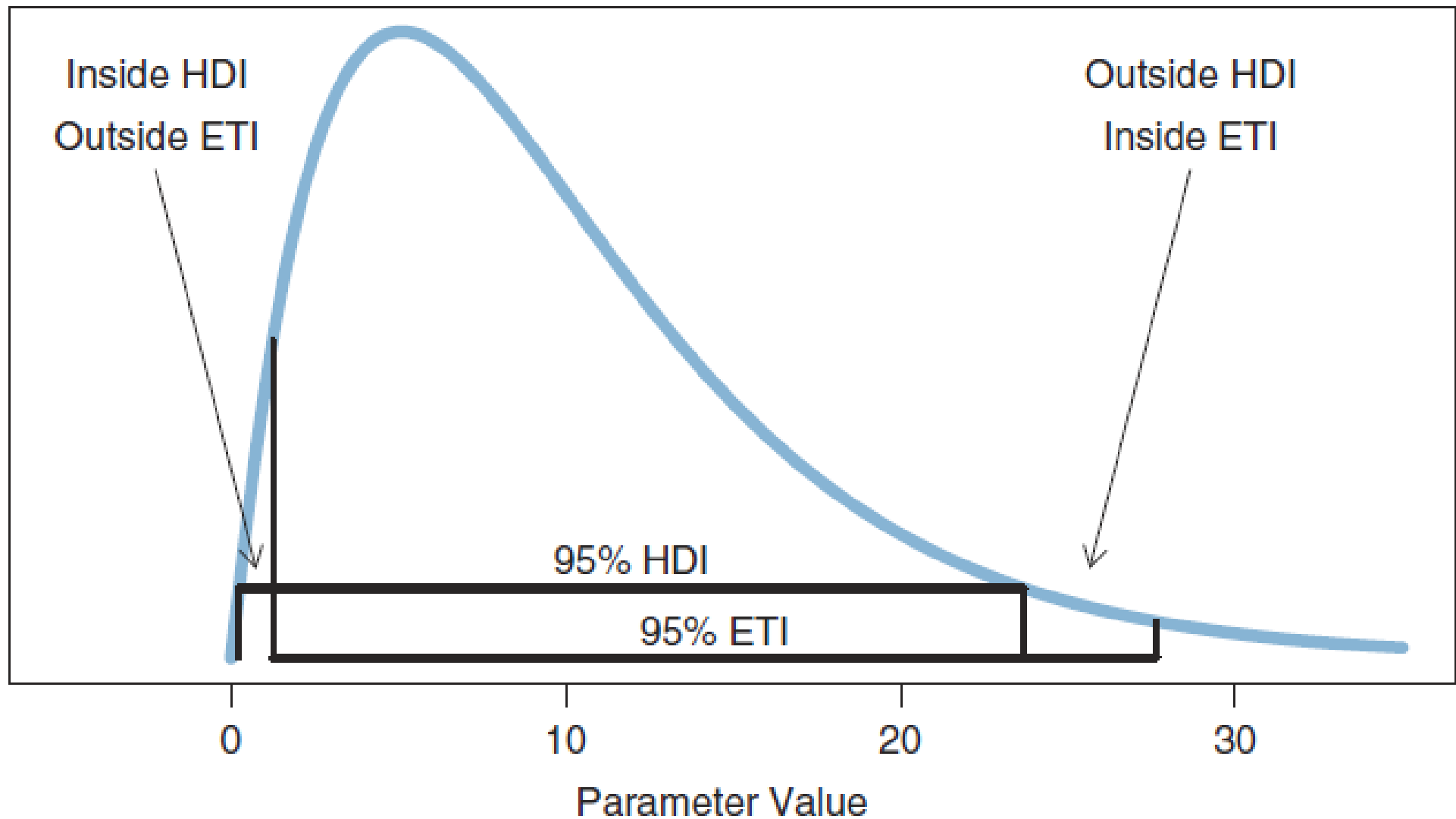


Figure 12.2 A skewed distribution has different 95% highest density interval (HDI) than 95% equal-tailed interval (ETI).

Example with Seven Islands

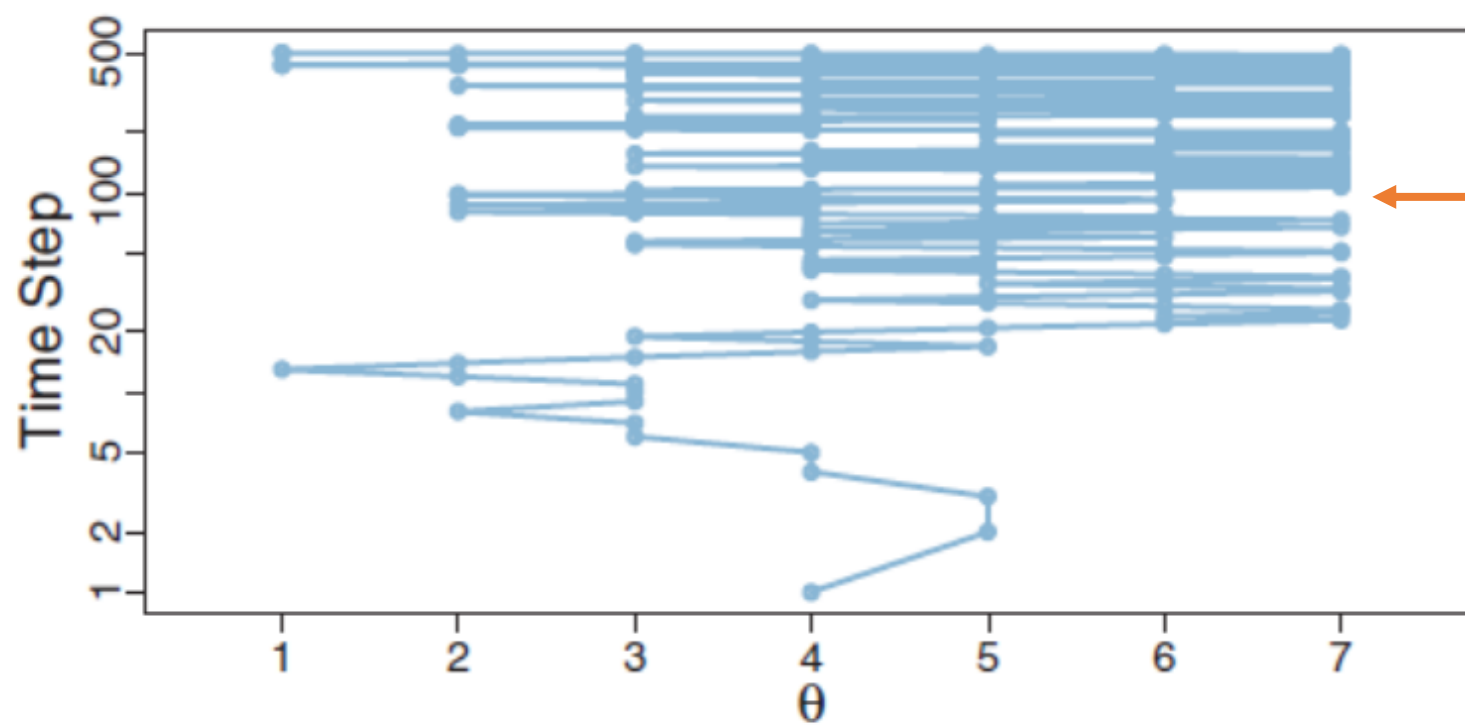
- Our goal is to get an accurate representation of the posterior distribution by sampling a large number of points from it. But how?
- Suppose a politician plans to travel from island to island. His goal is to visit all the islands proportionally to their relative population.
- Everyday he has to decide to (i) stay on the current, (ii) move to the west, or (iii) move to the east.

Entourage Capacity

- No idea what the total population of the island chain is, and doesn't even know exactly how many islands there are.
- They can ask the mayor of the island they are on how many people are on the island.
- When the politician proposes to visit an adjacent island, they can ask the adjacent mayor how many people are on that island.

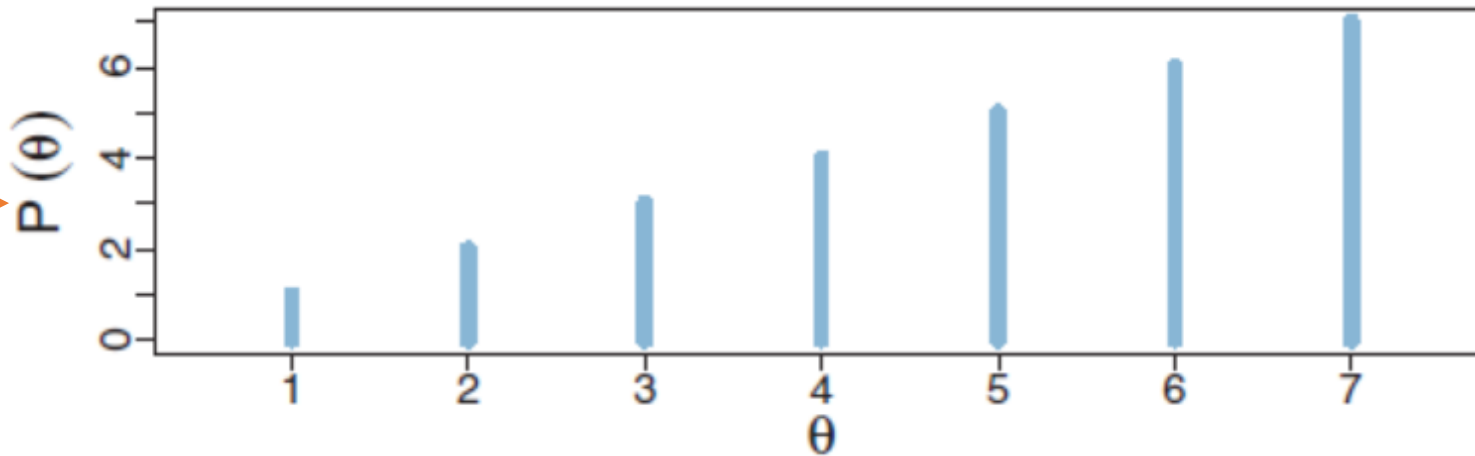
Metropolis Heuristic

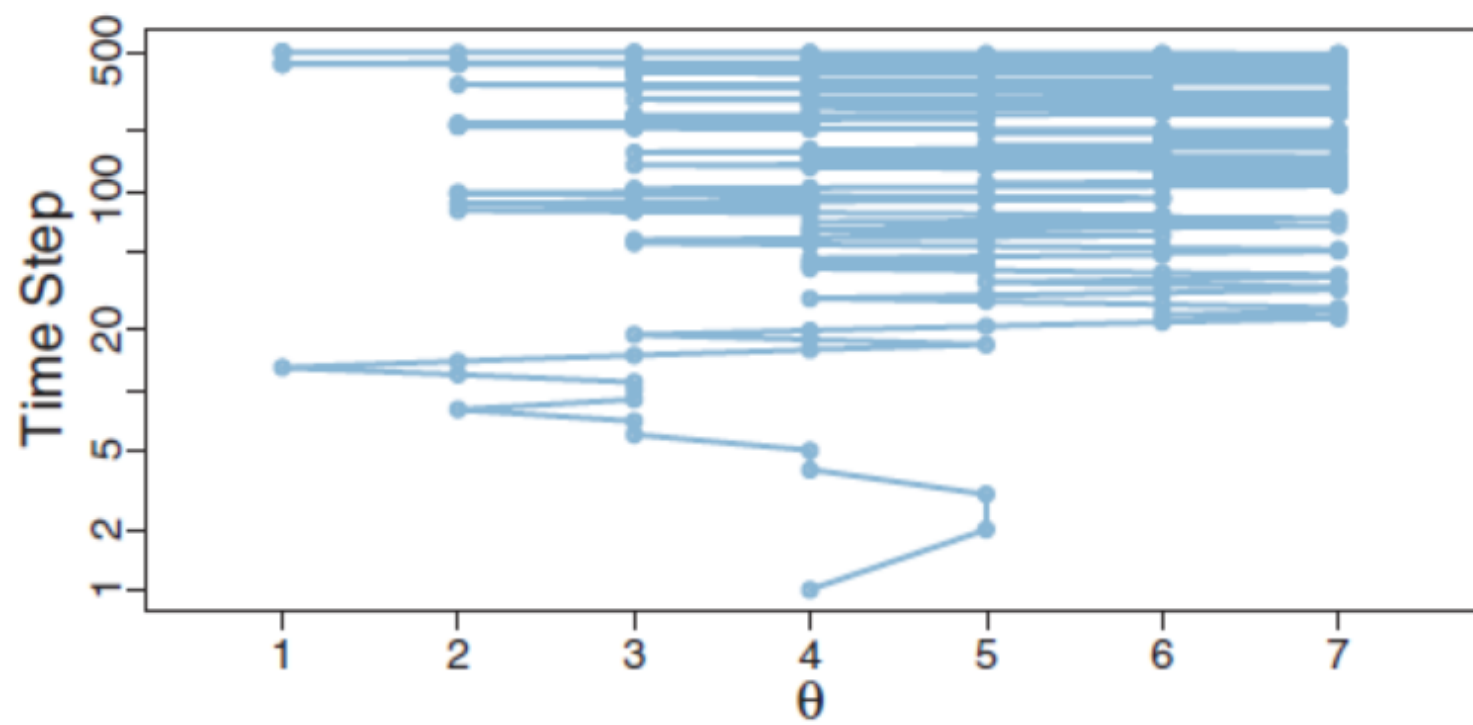
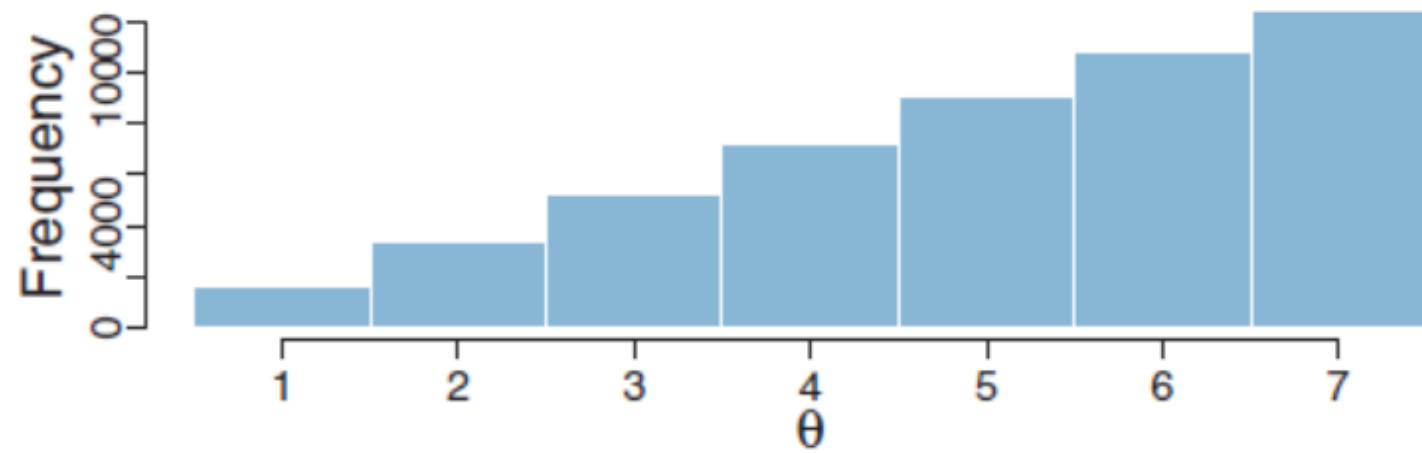
- Flip a coin to decide whether to propose the adjacent island to the east or to the west.
- If the proposed island has a larger population than the current one, he definitely goes to that island.
- If not, goes to the proposed island with a probability
$$p_{\text{move}} = P_{\text{proposed}} / P_{\text{current}}$$
 P_{proposed} (P_{current}) denotes the population of proposed (current) island (how to code?)



A **Markov chain** is a stochastic process where the probability of each event depends only on the exactly previous state.

We **don't** need to know the function of $p(\theta)$ here





General Properties

- Under some assumptions, given large enough time steps, the domain can be fully reached.
- Regardless of the double randomness, in the long run the relative frequency of visits mimics the target distribution.
- We combine the two probabilities to have:
$$p_{\text{move}} = \min \left(\frac{P(\theta_{\text{proposed}})}{P(\theta_{\text{current}})}, 1 \right)$$
where $P(\theta)$ does not need to be normalized.

“Only Ratio Matters” Helps!

Not the target
distribution



1. Recall that we need to sample from a proposal distribution to have θ_{proposed} .
 2. Then need to compute $P(\theta_{\text{proposed}})/P(\theta_{\text{current}})$.
 3. Finally need to generate a random value from a uniform distribution, to decide accept or reject θ_{proposed} .
- By being able to do the above 3 things, we can generate random samples from the target distribution *even when the target is not normalized*.

Denominator is Skipped

- Notice that the numerator of Bayes' rule $p(D|\theta)p(\theta)$ is exactly an *unnormalized posterior* distribution.
- By evaluating $p(D|\theta)p(\theta)$ with the above Metropolis algorithm, we can generate random representative samples from the posterior distribution *without normalizing it by $p(D)$* .
- That is, we skipped the most difficult aspect of Bayesian inference.

It has only been with the development of MCMC algorithms and software that Bayesian inference is applicable to complex data analysis, and it has only been with the production of fast and cheap computer hardware that Bayesian inference is accessible to a wide audience.

Why Metropolis Works?

1. Why the relative transition probabilities between adjacent positions exactly match the relative values of the target distribution?
- The ratio of the transition probabilities is

$$\begin{aligned}\frac{p(\theta \rightarrow \theta + 1)}{p(\theta + 1 \rightarrow \theta)} &= \frac{0.5 \min (P(\theta + 1)/P(\theta), 1)}{0.5 \min (P(\theta)/P(\theta + 1), 1)} \\ &= \begin{cases} \frac{1}{P(\theta)/P(\theta + 1)} & \text{if } P(\theta + 1) > P(\theta) \\ \frac{P(\theta + 1)/P(\theta)}{1} & \text{if } P(\theta + 1) < P(\theta) \end{cases} \\ &= \frac{P(\theta + 1)}{P(\theta)}\end{aligned}$$

Why Metropolis Works?

2. Why the approximation is stable? If the current probability of being in a position matches the target, then the Metropolis keeps it that way?
- The probability of moving to $\theta + 1$ (or $\theta - 1$) is $0.5 \min(P(\theta + 1)/P(\theta), 1)$ ($0.5 \min(P(\theta - 1)/P(\theta), 1)$).
 - The probability of staying at position θ is $0.5 \left[1 - \min\left(\frac{P(\theta + 1)}{P(\theta)}, 1\right) \right] + 0.5 \left[1 - \min\left(\frac{P(\theta - 1)}{P(\theta)}, 1\right) \right]$

Transition Probability Matrix

$$\begin{bmatrix}
 \ddots & 0.5\min\left(\frac{P(\theta-1)}{P(\theta-2)}, 1\right) & 0 & 0 & 0 \\
 \ddots & 0.5\left[1 - \min\left(\frac{P(\theta-2)}{P(\theta-1)}, 1\right)\right] & 0.5\min\left(\frac{P(\theta)}{P(\theta-1)}, 1\right) & 0 & 0 \\
 & + 0.5\left[1 - \min\left(\frac{P(\theta)}{P(\theta-1)}, 1\right)\right] & & & \\
 0 & 0.5\min\left(\frac{P(\theta-1)}{P(\theta)}, 1\right) & 0.5\left[1 - \min\left(\frac{P(\theta-1)}{P(\theta)}, 1\right)\right] & 0.5\min\left(\frac{P(\theta+1)}{P(\theta)}, 1\right) & 0 \\
 & & + 0.5\left[1 - \min\left(\frac{P(\theta+1)}{P(\theta)}, 1\right)\right] & & \\
 0 & 0 & 0.5\min\left(\frac{P(\theta)}{P(\theta+1)}, 1\right) & 0.5\left[1 - \min\left(\frac{P(\theta)}{P(\theta+1)}, 1\right)\right] & \ddots \\
 & & & + 0.5\left[1 - \min\left(\frac{P(\theta+2)}{P(\theta+1)}, 1\right)\right] & \\
 0 & 0 & 0 & 0.5\min\left(\frac{P(\theta+1)}{P(\theta+2)}, 1\right) & \ddots
 \end{bmatrix}$$

Matrix Multiplication

- We can denote $w = (w_1, w_2, w_3, \dots, w_R)$ as a row vector of the current location probabilities.
- When $w = (\dots, 0, 1, 0, \dots)$ with a 1 only in the θ position, wT is simply the θ th row of T .
- When $w = (\dots, 0, w_3, w_4, 0, \dots)$, wT is a weighted sum of two rows $w_3T_3 + w_4T_4$. (`np.dot()`)
- Therefore, wT yields the position probabilities for the next time step.

Proof of Stability

- Suppose the current position probabilities match the target probabilities, i.e., $w = (\dots, P(\theta - 1), P(\theta), P(\theta + 1), \dots)/Z$, where $Z = \sum_{\theta} P(\theta)$ is the normalizer.
- You can see that the θ th component of wT is

$$\begin{aligned} \sum_r w_r T_{r\theta} = & P(\theta - 1)/Z \cdot 0.5 \min\left(\frac{P(\theta)}{P(\theta - 1)}, 1\right) \\ & + P(\theta)/Z \cdot \left(0.5 \left[1 - \min\left(\frac{P(\theta - 1)}{P(\theta)}, 1\right)\right] + 0.5 \left[1 - \min\left(\frac{P(\theta + 1)}{P(\theta)}, 1\right)\right]\right) \\ & + P(\theta + 1)/Z \cdot 0.5 \min\left(\frac{P(\theta)}{P(\theta + 1)}, 1\right) \end{aligned} \tag{7.4}$$

Proof of Stability

- Four cases need to be analyzed:
 - (i) $P(\theta) > P(\theta - 1)$ and $P(\theta) > P(\theta + 1)$
 - (ii) $P(\theta) > P(\theta - 1)$ and $P(\theta) < P(\theta + 1)$
 - (iii) $P(\theta) < P(\theta - 1)$ and $P(\theta) > P(\theta + 1)$
 - (iv) $P(\theta) < P(\theta - 1)$ and $P(\theta) < P(\theta + 1)$
- Working through the 4 cases, we find that it always reduces to $P(\theta)/Z$.
- The θ th component starts at $P(\theta)/Z$, then it stays at $P(\theta)/Z$.

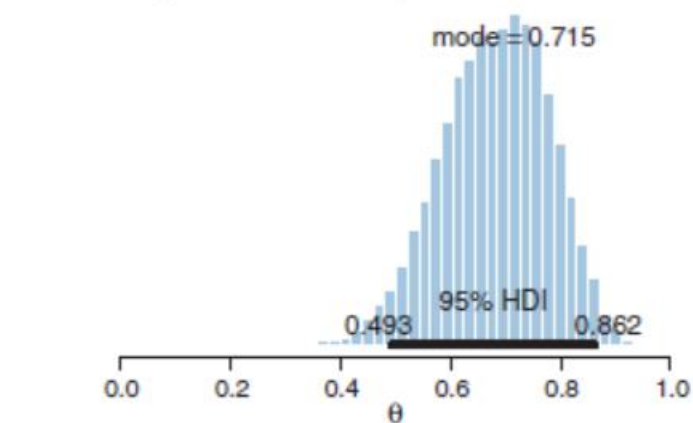
Proof of Stability

- (i) $P(\theta) > P(\theta - 1)$ and $P(\theta) > P(\theta + 1)$

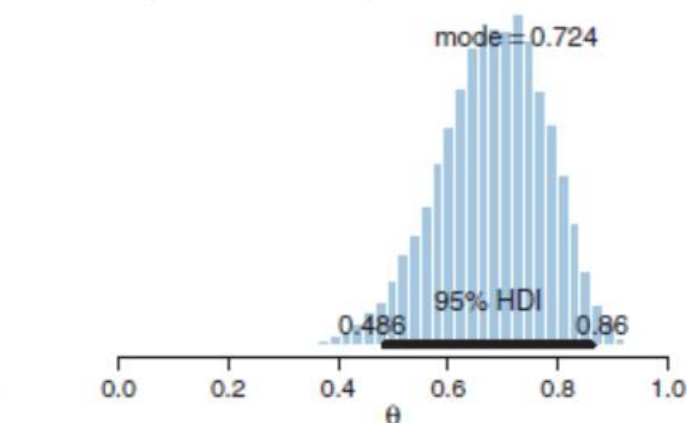
Equation (7.4) becomes

$$\begin{aligned}\sum_r w_r T_{r\theta} &= P(\theta - 1)/Z \cdot 0.5 \\ &\quad + P(\theta)/Z \cdot \left(0.5 \left[1 - \left(\frac{P(\theta - 1)}{P(\theta)} \right) \right] + 0.5 \left[1 - \left(\frac{P(\theta + 1)}{P(\theta)} \right) \right] \right) \\ &\quad + P(\theta + 1)/Z \cdot 0.5 \\ &= 0.5 P(\theta - 1)/Z \\ &\quad + 0.5 P(\theta)/Z - 0.5 P(\theta)/Z \frac{P(\theta - 1)}{P(\theta)} + 0.5 P(\theta)/Z - 0.5 P(\theta)/Z \frac{P(\theta + 1)}{P(\theta)} \\ &\quad + 0.5 P(\theta + 1)/Z \\ &= P(\theta)/Z\end{aligned}$$

Prpsl.SD = 0.02, Eff.Sz. = 468.9



Prpsl.SD = 0.2, Eff.Sz. = 11723.9



Prpsl.SD = 2, Eff.Sz. = 2113.4

